

Methods of proof – 3

The extremal principle and compactness

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Theoretical minimum.

Fact 1 (take the extreme one). On a finite set a maximal element always exists, and the whole proof consists in choosing what to maximise: the longest segment, the triangle of largest area, the vertex of highest degree, the counterexample with the smallest number of elements. The chosen extremal object then satisfies an equation — “nothing can be improved” — and that equation is the content of the theorem.

Theorem 1 (compactness replaces finiteness). On an infinite set the maximum need not exist, and the substitutes are: a continuous function on a compact set attains its maximum (Continuous functions – 7); a bounded sequence has a convergent subsequence; a uniformly bounded equicontinuous family of functions has a uniformly convergent subsequence (Arzelà–Ascoli, Functional sequences, series and spaces – 1); a closed bounded convex set has extreme points (Optimization – 2).

Theorem 2 (the variational method). To prove that an object with a property exists, maximise a functional over a compact family and show that the maximiser has that property, because otherwise it could be improved. This is how one obtains John’s ellipsoid (Convex – 8), the best approximation (Continuous functions – 9), the eigenvector of the largest eigenvalue by the Rayleigh quotient (Linear algebra – 7) and the isoperimetric inequality (Convex – 4).

Fact 2 (when the extremum does not exist). If the supremum is not attained, either the family is not compact — then one compactifies, or passes to a limit and proves that the limit is admissible — or the statement is false. Checking the existence of the extremum is a separate step of the proof and is exactly where most wrong solutions break.

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1. Prove by the extremal principle: among finitely many points in the plane, not all collinear, there is a line containing exactly two of them (Sylvester–Gallai); and in a finite set of points the two closest ones have the property that no third point lies inside their lens.
 2. Prove that a continuous function on a compact set attains its maximum, and that the maximum of $\frac{\langle Ax, x \rangle}{\langle x, x \rangle}$ over the unit sphere is the largest eigenvalue of a symmetric A ; deduce the existence of an orthogonal eigenbasis by induction on the dimension.
 3. Let K be a convex body in $\mathbb{R}^n$. Prove that among all ellipsoids contained in K there is one of maximal volume, that it is unique, and state what the contact points give (the full statement is in Convex – 8).
 4. Prove that every continuous $f: [0, 1] \rightarrow [0, 1]$ has a fixed point by considering the supremum of $\{x : f(x) \geq x\}$, and explain why the same argument fails for a discontinuous function and for a map of the open interval.
 5. Prove that a family of closed sets in a compact space with the finite intersection property has a common point, and deduce that an infinite graph all of whose finite subgraphs are k -colourable is itself k -colourable (De Bruijn–Erdős); identify the place where compactness replaces the extremal principle.